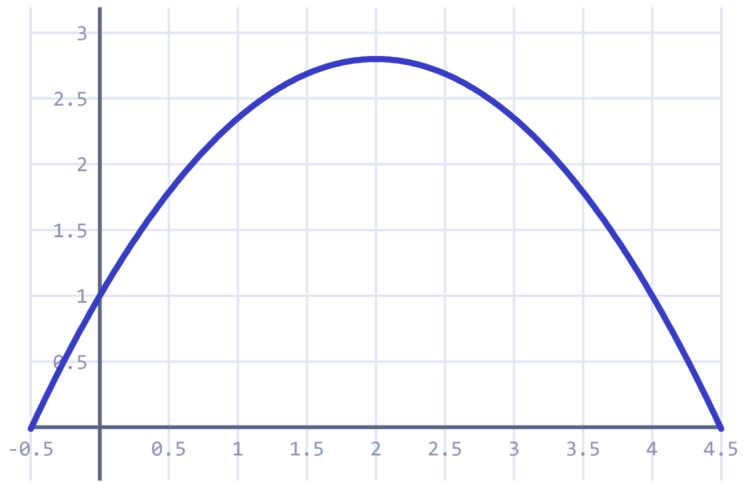
Why does everything you throw come back down — in a curve?

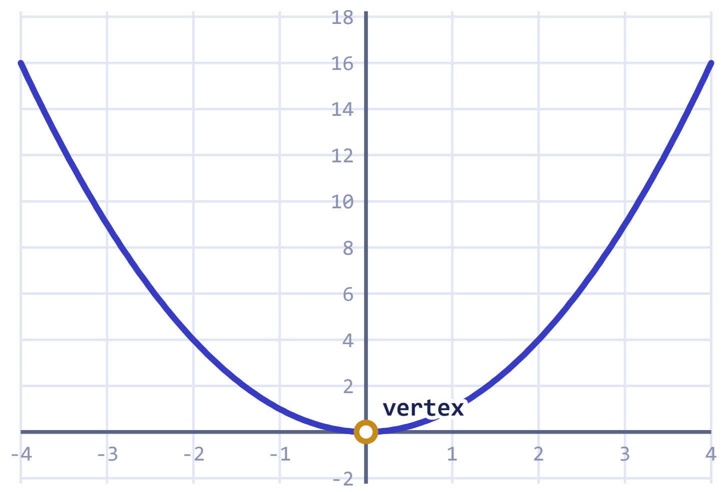
- A thrown ball, a fountain, a bridge cable, a profit model — same shape.
- That shape is a **parabola**, and this week we learn to read it and build it.
- By Friday you'll graph one from an equation and find its highest point.



height vs. time of a thrown ball

The quadratic function

- Standard form: $f(x) = ax^2 + bx + c$, with $a \neq 0$.
- The graph is always a **parabola**.
- $a > 0$ opens up (a valley); $a < 0$ opens down (a hill).
- $|a|$ controls how narrow or wide it is.



the parent graph y = x²

Graph f(x) = x² − 4x + 3

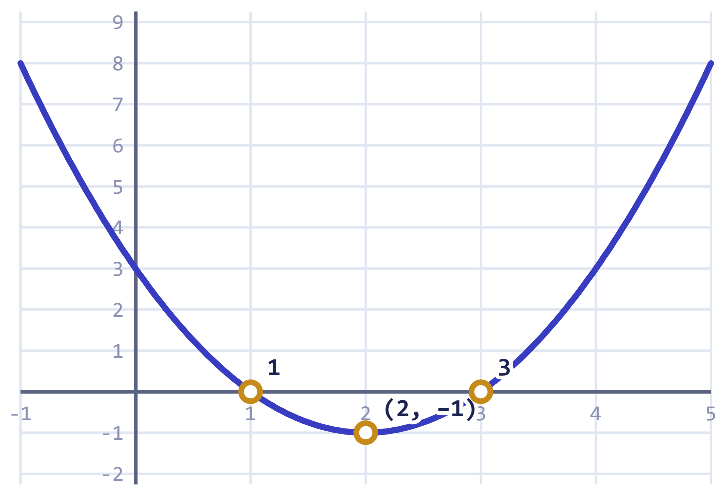
WORKED EXAMPLE

1. Axis of symmetry: $x = -\dfrac{b}{2a} = -\dfrac{-4}{2} = 2.$

2. Vertex y: $f(2) = 4 - 8 + 3 = -1 \Rightarrow (2, -1).$

3. Zeros: $x^2 - 4x + 3 = (x-1)(x-3) = 0 \Rightarrow x = 1, 3.$

4. y-intercept: $f(0) = 3.$



vertex (2, −1) · roots 1 and 3

WATCH OUT

The one sign error everyone makes

COMMON MISTAKE

The axis of symmetry is $x = -\dfrac{b}{2a}$ — that leading **minus** is part of the formula.

For $f(x) = x^2 - 4x + 3$, $b = -4$, so $x = -\dfrac{-4}{2} = +2$, **not** -2.

KEEP IN MIND

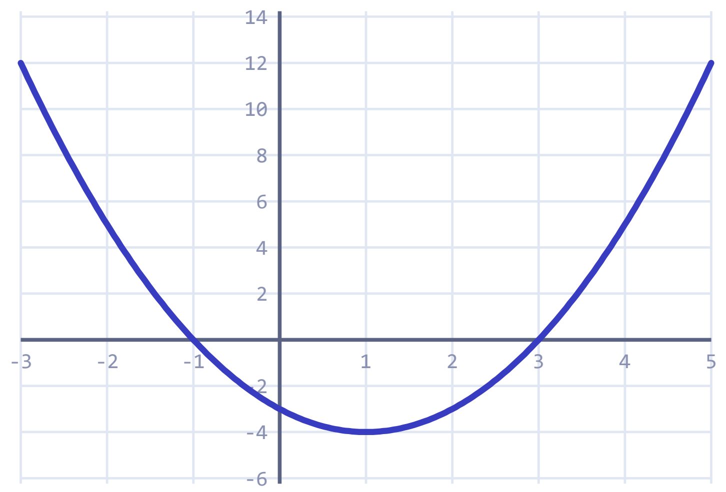
Write the negative first, then substitute b. Two negatives make the positive.

Your turn

YOU TRY

Graph $f(x) = x^2 - 2x - 3$.

Find: (a) the axis of symmetry, (b) the vertex, (c) the zeros.



sketch it, then check against the screen

VOCABULARY

Words we'll use all unit

Parabola — the U-shaped graph of a quadratic

Vertex — the highest or lowest point

Axis of symmetry — the vertical line through the vertex

Zeros / roots — where the graph crosses the x-axis

Standard form — $f(x) = ax^2 + bx + c$

Where we're headed this week

- **Mon** — graph from standard form (today).
- **Tue** — vertex form and transformations.
- **Wed** — solve by factoring & the quadratic formula.
- **Thu** — real-world max/min (projectile & profit).
- **Fri** — quiz + a modeling challenge.